

Assignment -II

Power Station Engineering (PSE), 8th Sem., Electrical Engineering Module-4 (Thermal Power Plant)

1. List the factors to be considered while choosing a site for thermal power plant.
2. Name the four major circuits in steam power plant.
3. What consists of air and flue gas circuit?
4. What consists of feed water and steam flow circuit in thermal power plant?
5. What consists of cooling water circuit and coal & ash circuit in thermal power plant?
6. Why the preparation of coal is necessary?
7. How the ash handling system is classified?
8. Why ash handling system is needed?
9. Name different types of dust collectors?
10. What are the uses of ash and dust produced from the thermal power plant?
11. What is the main purpose of chimney?
12. What is the difference between stocker firing and pulverised fuel firing?
13. State the advantage of pulverised fuel firing.
14. What is pulverization?
15. What is the main purpose of high-pressure boilers?
16. What is the use of air preheater?
17. Define draught, what is the use of draught in thermal power plants?
18. What is the necessity of Automatic controls for feed water?
19. What do you understand by the term "Boiler draught"?
20. What is the purpose of automatic combustion control?
21. Define super heater.
22. What is the function of economizer in thermal power plants?
23. What is the inherent demerit of the technique used in economizer of thermal power plant?
24. Explain the term Regeneration?
25. Explain Reheat cycle?
26. What is bleeding of steam in a thermal power plant?
27. What is meant by cooling Towers?
28. What is the working principle of Cooling Towers?
29. What is Deaeration?
30. What is the purpose of Deaeration?

31. Why thermal power plants are not suitable for supplying peak loads?
32. Give the general layout of ash handling and dust collection system of a thermal power station. Explain it briefly.
33. Draw the general layout of thermal power plant and explain the working of different circuits.
34. What is condenser? Name the different types of condenser. Describe the operation of condenser.
35. How are dust collectors classified? Explain the operation of an electrostatic Precipitator in a thermal Power Plant.
36. Explain the working principle of Cooling Tower and Spray pond of the thermal Power Plant.
37. What are the basic differences between water tube and fire tube boilers of a thermal power plant?
38. Write the advantages and disadvantages of water tube boiler over fire tube boiler.
39. Discuss the necessity of superheated steam, Pulverized coal and preheated air in thermal power plant.
40. What is governing system? Name the different types of governing system. Describe the operation of governing system.
41. State the advantages and disadvantages of a thermal power plant as compared to hydroelectric power plant.